Unit 6.5: Cubic & Cube Root Functions Name: ____________________

Translations (Shifts)

The Big Idea

Translations **slide** the graph without changing its shape. The "S" curve stays the same — it just moves to a new location.

Cubic Form	Cube Root Form	What moves?
$f(x) = (x-h)^3 + k$	$f(x) = \sqrt[3]{x-h} + k$	Center point goes to (h, k)

How to Find h and k

	What it does	Watch out!
h	Shifts **left/right** $h > 0$: right $h < 0$: left	$(x-3)$ means $h = 3$ (right) $(x+2)$ means $h = -2$ (left)
k	Shifts **up/down** $k > 0$: up $k < 0$: down	This one is straightforward! $+5$ means up 5, -3 means down 3

Transformation Cheat Note: To move your points, use this rule:

$$(x, y) \longrightarrow (x + h,\ ay + k)$$

(Note: For today's lesson, $a = 1$)

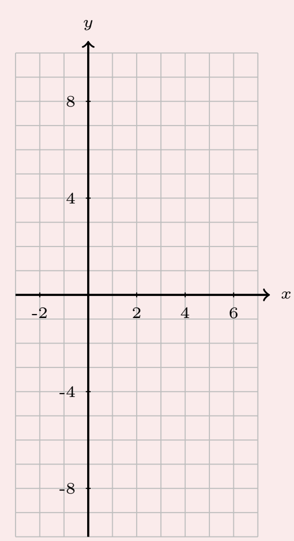
Example 1: $f(x) = (x-2)^3 + 1$ (Cubic)

Step 1: Find h and k
$h =$ ______ $k =$ ______

Step 2: Find the center point
Center: (,)

Step 3: Shift the anchor points
Apply $(x + h, y + k)$

Parent	Shifted
$(-2, -8)$	(,)
$(-1, -1)$	(,)
$(0, 0)$	(,)
$(1, 1)$	(,)
$(2, 8)$	(,)



Example 2: $f(x) = \sqrt[3]{x+3} - 2$ (Cube Root)

Step 1: Find h and k

$h =$ _______ $k =$ _______

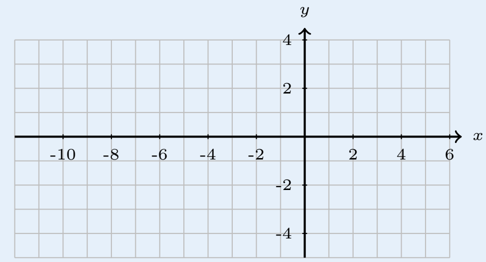
Step 2: Find the center point

Center: (,)

Step 3: Shift the anchor points

Apply $(x+h, y+k)$

Parent	Shifted
$(-8,-2)$	(,)
$(-1,-1)$	(,)
$(0,0)$	(,)
$(1,1)$	(,)
$(8,2)$	(,)



Anchor Points to Memorize

Cubic: $f(x) = x^3$	**Cube Root:** $f(x) = \sqrt[3]{x}$
$(-2,-8), (-1,-1), (0,0), (1,1), (2,8)$	$(-8,-2), (-1,-1), (0,0), (1,1), (8,2)$

To shift: Add h to every x-coordinate. Add k to every y-coordinate.